

David Henderson

Software Engineer | Data Scientist

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San Francisco Bay Area

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SUMMARY

Software Engineer and AWS-certified Machine Learning Specialist with 6+ years of experience designing scalable software systems, building data pipelines, and optimizing database performance. Strong background in backend development, distributed systems, and cross-functional collaboration. Proven track record of delivering production-quality code, contributing to open-source, and leading technical projects.

TECHNICAL SKILLS

- **Programming & Data Analysis:** Python, R, SQL, SAS, Tableau, Linux, PowerBI
- **Machine Learning & Statistical Tools:** PyTorch, TensorFlow/Keras, MLflow, Predictive Modeling, Statistical Analysis
- **Cloud Computing:** AWS Certified Machine Learning – Specialty (2025), AWS Certified AI Practitioner (2024)
- **Data Management & Visualization:** ETL processes, data transformation, visualization tools, GIS

EXPERIENCE

Software Developer

2024

Galactic Advisors

- As part of the software development and quality control teams, utilized strict version control (Git) and development processes.
- Developed protocols for quality control. Maintained and created multi-structured database features. Optimized query rules for data transfer, analyses and automatic processes including data upload, preprocessing, formatting, analysis, and report generation.
- Updated most critical, most used sections of database schema within first 30 days.
- Collaborated with diverse teams (security, software, compliance) under tight deadlines to meet project deliverables.

Research Biologist (Ph.D.)

2018—2023

Department of Biology, Washington University in St. Louis

- Conducted machine learning and Species Distribution Modeling analyses to determine methods to utilize less precise data resulting in enhanced functionality for the enmSdmX package in R ([link](#)).
- Spearheaded projects related to large scale network analyses using new and existing tools to interpret, analyze, and visualize multivariate relationships between hosts and pathogens
- Led 3 projects making use of bioinformatic, environmental, and multiple high-dimensional data types, developing methods to measure the impact of pathogens/natural enemies on host organisms.
- Co-organized and led an Andes-Biodiversity working group in English and Spanish with collaborators from South America, Europe and the United States that set the foundation for the future projects.

Visiting Faculty

2022

Dr. Brian Sedio Lab, University of Texas at Austin

- Found solutions to update previous bioinformatic pipelines and created original modeling approaches to define and compare chemical profiles (consisting of thousands of unique compounds) among individuals or groups.
- Gained familiarity with the manipulation and analysis of high dimensional data and implementing ML models in scalable formats.
- Communicated with faculty and technicians to understand and perform analytic tasks including: contributions to the creation of training manuals, documentation for lab protocols and procedures, and data pipelines.

Biotech Consultant

2021—2023

The Biotechnology and Life Science Advising (BALSA) Group

- Worked as a project-based consultant, as a team member of a non-profit, student-led consulting group.
- Selected appropriate statistical tools to address specific client needs. Summarized findings in written and oral forms.
- Performed primary and secondary research including Surveys and Interviews with KPIs, Market Segmentation, Market Sizing, Competitive Analyses (SWOT), and Exit Strategies research for academic inventors, biotech innovators and investors, in a team-based environment.
- Stayed abreast of the latest developments and innovations, and stayed current on regulations and standards relevant to the project

Research Program Mentor

2023—Present

Polygence

- Guide high school students through self-directed research projects involving data analysis, experimental design, and scientific writing.
- Mentor projects applying statistical modeling and environmental science to real-world challenges.
- Foster critical thinking by teaching students to interpret and present analytical findings effectively.

EDUCATION

Washington University in St. Louis

2018—2023

Ph.D., Biology and Biomedical Sciences

Northern Illinois University - Dekalb

2014—2016

B.Sc., Biological Studies – *Summa Cum Laude*